05 February 2024 10:58

3) Associativity of convolution

$\{x(n)\}$, $\{h_1(n)\}$, $\{h_2(n)\}$ } are absolutely summable DT signals

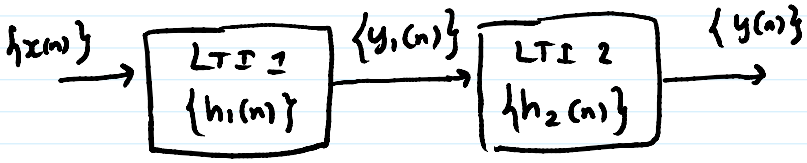
LHS

$$\left(\{x(n)\} * \{h_1(n)\}\right) * \{h_2(n)\} = \{x(n)\} * \left(\{h_1(n)\} * \{h_2(n)\}\right)$$

RHS

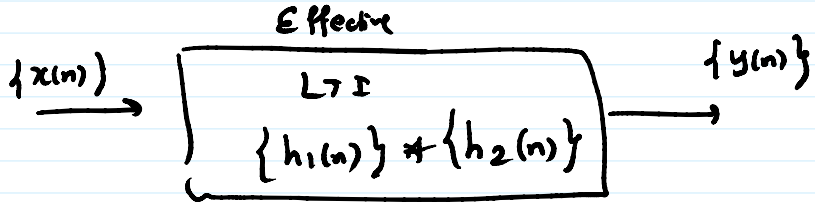
$\{y_1(n)\}$

LHS

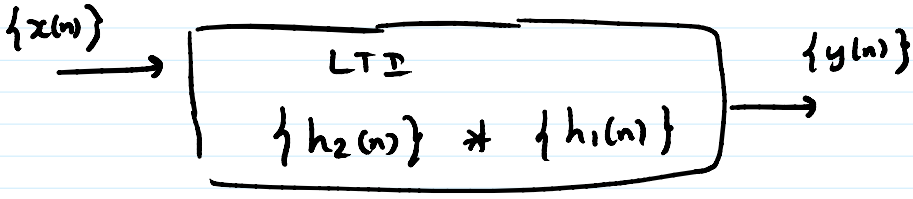


⇕ Equivalent

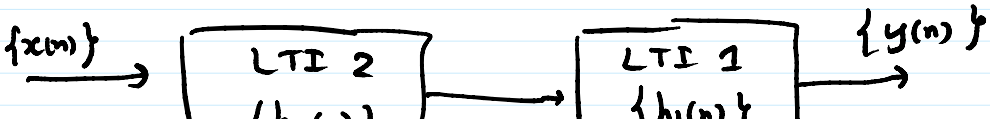
RHS

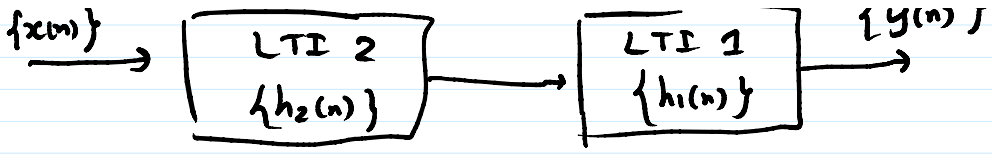


⇕ Equivalent (Commutativity)



⇕ Equivalent (associativity)





Order of LTI systems in a cascade can be interchanged.

4. Distributivity

$$\{x(n)\} * \left(\{h_1(n)\} + \{h_2(n)\} \right)$$

$$= \{x(n)\} * \{h_1(n)\} + \{x(n)\} * \{h_2(n)\}$$



Connections between impulse responses of LTI systems and their properties.

1) Causal LTI Systems.

$$\{y(n)\} = \{x(n)\} * \{h(n)\}$$

$$= \{h(n)\} * \{x(n)\}$$

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)\, x(n-k)$$

$$y(n) = \underbrace{\cdots + h(-1)x(n+1)}_{k=-2,\ k=-1} + \underbrace{h(0)x(n) + h(1)x(n-1) + \cdots}_{k=0,\ k=1,\ k=2}$$

future inputs.

Present & past inputs

If $h(n) = 0$ for all $n < 0$

then $y(n)$ does not depend on future samples.

Causality Condition

If $h(n) = 0$ for all $n \neq 0$

then LTI system is memoryless.

x ——— x

Stable LTI systems.

If impulse response $\{h(n)\}$ is absolutely summable, then LTI system is stable

(Necessary & Sufficient)

Proof: (Sufficiency)

Input Bounded $|x(n)| < B_x$ for all n.

$\Downarrow$ finite const.

∞

"finite const.

$$\sum_{n=-\infty}^{\infty} |h(n)| = C < \infty$$

$$|y(n)| = \left| \sum_k h(k)\, x(n-k) \right|$$

$|A+B| \le |A| + |B|$

$$\le \sum_k |h(k)\, x(n-k)|$$

$$|h(k)|\, |x(n-k)|$$

$|x(n-k)| < B_x$

$$\le \sum_k B_x |h(k)|$$

$$\le B_x \underbrace{\sum_k |h(k)|}_{C}$$

for all n: $|y(n)| \le B_x C$.

$\Rightarrow$ Bounded output $\Rightarrow$ stable.

Proof for "Necessary" Part.

- If the impulse response $\{h(n)\}$ is not absolutely summable, then sys. is unstable

We have $\sum_n |h(n)| = \infty$ → conjugate

Consider input $x(n) = \dfrac{h^*(-n)}{|h(-n)|}$

$|x(n)| = 1$ for all n

Bounded input.

Output at $n=0$

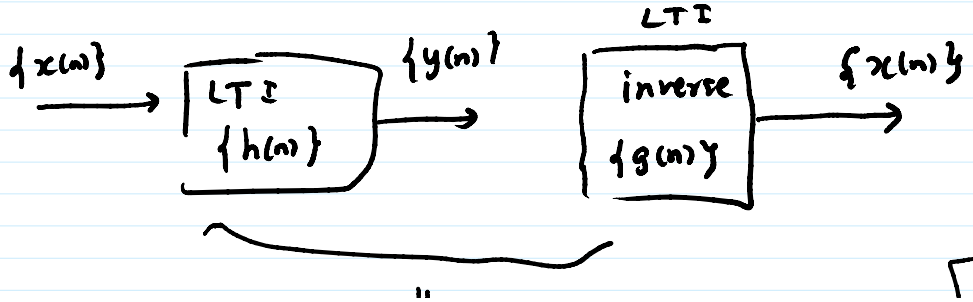
$$y(0) = \sum_k h(k)\, x(0-k)$$

$$x(0-k) = \frac{h^*(k)}{|h(k)|}$$

$$= \sum_k \frac{|h(k)|^2}{|h(k)|}$$

$$= \sum_k |h(k)| = \infty$$

Invertibility of LTI systems



$$\{h(n)\} * \{g(n)\} = \{\delta(n)\}$$

Requirement to find inverse.

length N input
length L impulse resp
length of output
N+L-1